

Network Layer: Addressing



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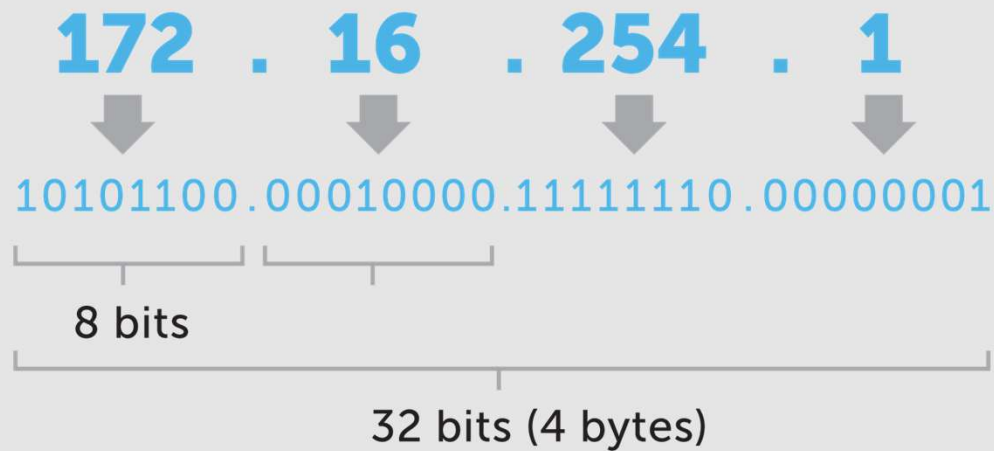
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Outline

IPv4 Address Space

Classless and Classful Addressing

IPv4 address in dotted-decimal notation



- ❖ An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

- ❖ The address space of IPv4 is 2^{32} or 4,294,967,296.

IPv4

Dotted-Decimal notation and Binary Notation

The diagram illustrates the conversion of a 32-bit binary address to dotted-decimal notation. The binary address is divided into four 8-bit segments: 10000000, 00001011, 00000011, and 00011111. Lines connect each segment to its corresponding decimal value in the dotted-decimal notation 128.11.3.31. The first segment (10000000) connects to 128, the second (00001011) to 11, the third (00000011) to 3, and the fourth (00011111) to 31.

Binary Segment	Decimal Value
10000000	128
00001011	11
00000011	3
00011111	31

128.11.3.31

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

❖ Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

a. 129.11.11.239

b. 193.131.27.255

Classful Addressing

- In classful addressing, the address space is divided into five classes:
A, B, C, D, and E

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation



Example

❖ Find the class of each address.

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 14.23.120.8
- d. 252.5.15.111

Solution

- a. The first bit is 0. This is a class A address.
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.
- c. The first byte is 14; the class is A.
- d. The first byte is 252; the class is E.

Classful Address Block Size

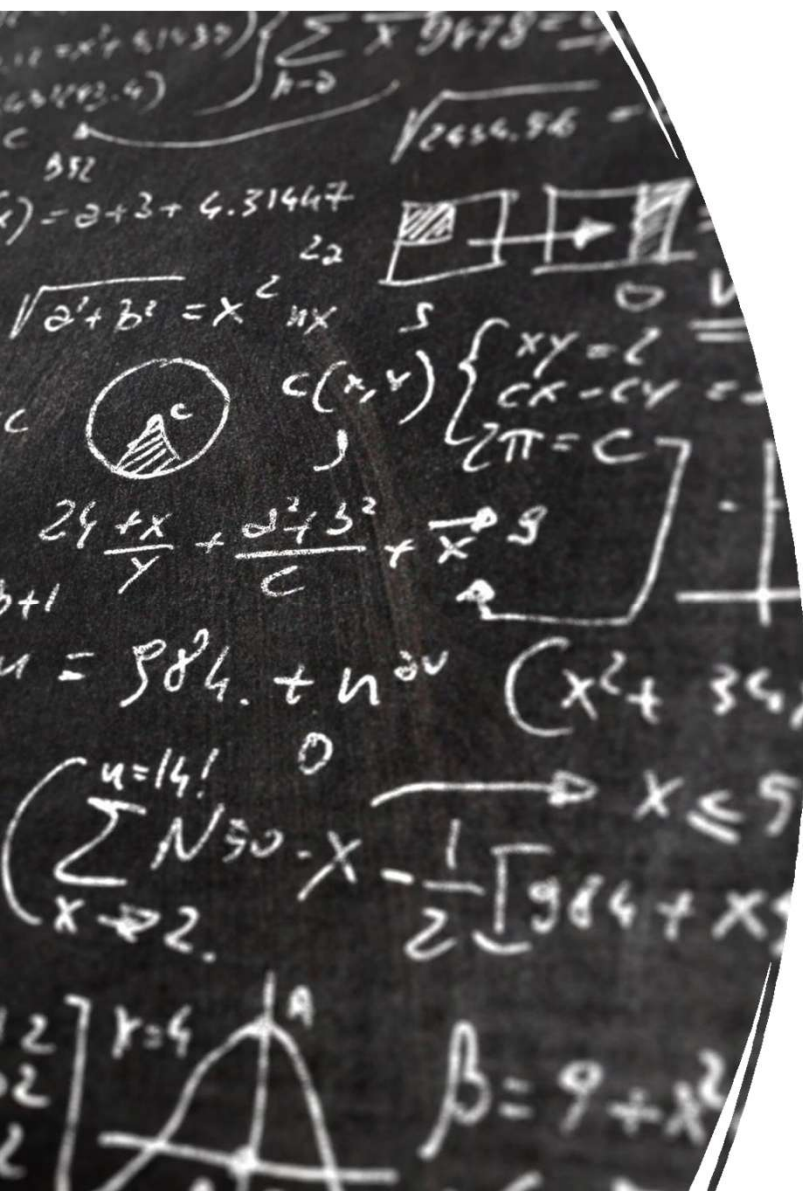
<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Default Masks for Classful Addressing

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

CIDR Notation

- ❖ In IPv4 addressing, a block of addresses can be defined as $x.y.z.t / n$ in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.
- ❖ The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.
- ❖ The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.



Example

* Same example is referenced in next three slides

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first and last address in the block?

Solution:

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000 or 205.16.37.32.

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111 or 205.16.37.47

First and Last Address: AND/OR Operations

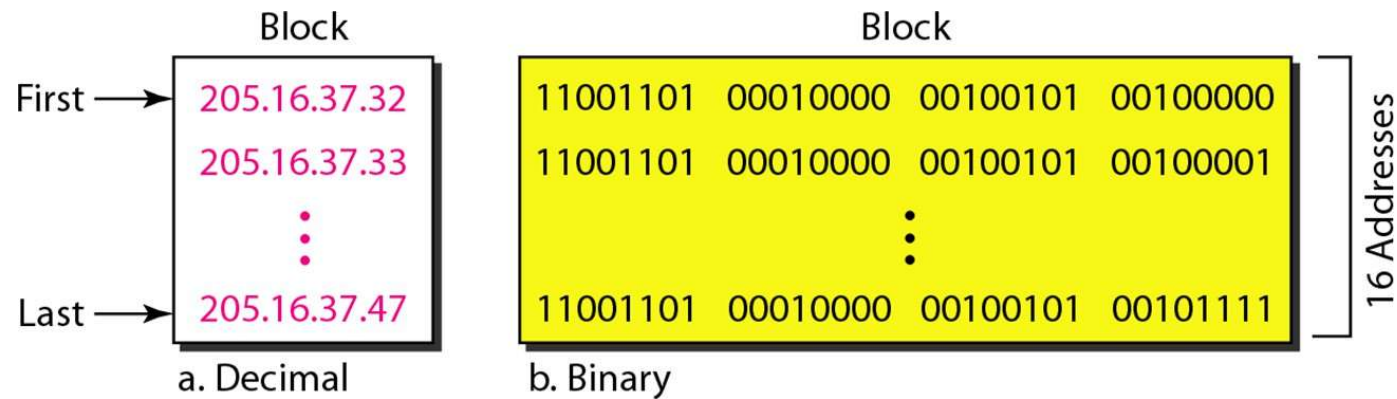
The first address can be found by ANDing the given addresses with the mask

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

The last address can be found by ORing the given addresses with the complement of the mask.

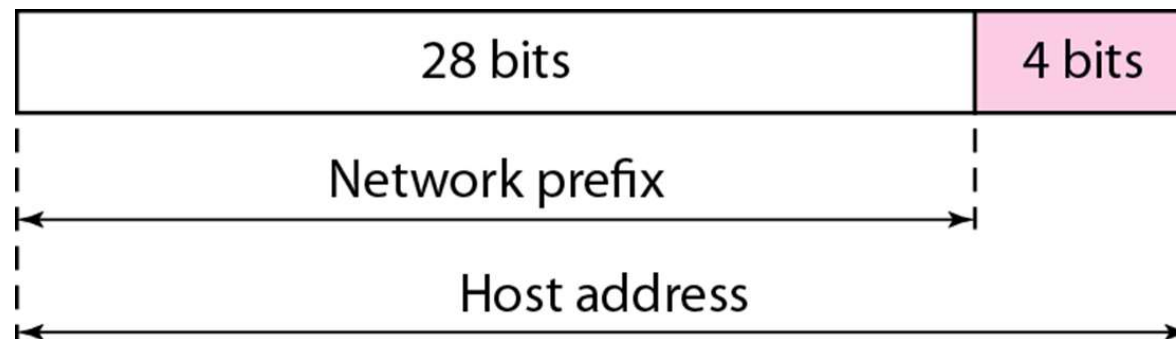
Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

Network configuration for the block 205.16.37.32/28



Note

- ❖ The number of addresses in the block can be found by using the formula 2^{32-n} .
- ❖ The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.
- ❖ Each address in the block can be considered as a two-level hierarchical structure: the leftmost n bits (prefix) define the network; the rightmost $32 - n$ bits define the host.



How Do You Get IPs?

- IP address ranges controlled by IANA



Internet Assigned Number Authority

Roots go back to 1972, ARPANET, UCLA

Today, part of ICANN

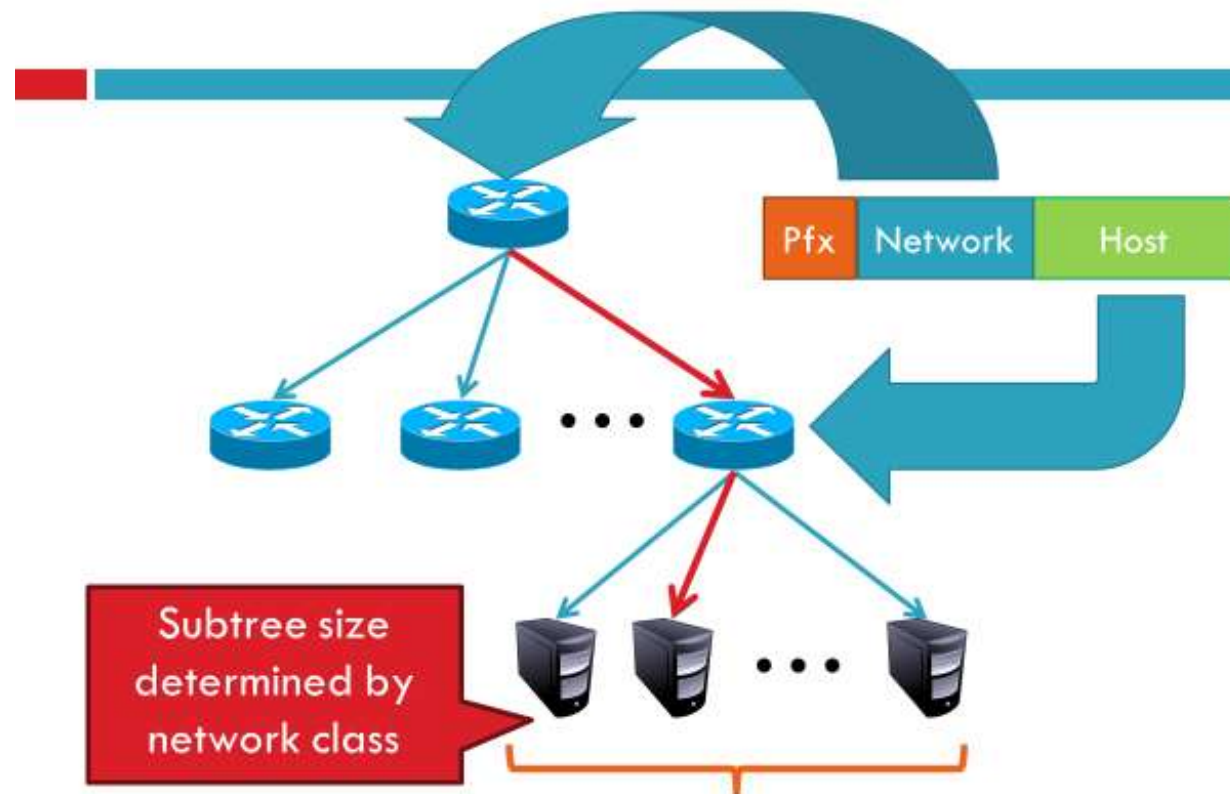
IANA grants IPs to regional authorities

ARIN (American Registry of Internet Numbers) may grant you a range of IPs

You may then advertise routes to your new IP range

There are now secondary markets, auctions, ...

Classless Addressing: The Two Level Hierarchy



Classless Addressing: Class Sizes and Issues

Class	Prefix Bits	Network Bits	Number of Classes	Hosts per Class
A	1	7	$2^7 - 2 = 126$ (0 and 127 are reserved)	$2^{24} - 2 = 16,777,214$ (All 0 and all 1 are reserved)
B	2	14	$2^{14} = 16,398$	$2^{16} - 2 = 65,534$ (All 0 and all 1 are reserved)
C	3	21	$2^{21} = 2,097,152$	$2^8 - 2 = 254$ (All 0 and all 1 are reserved)
Total: 2,114,036				

Way too big

Too many network IDs

Too small to be useful

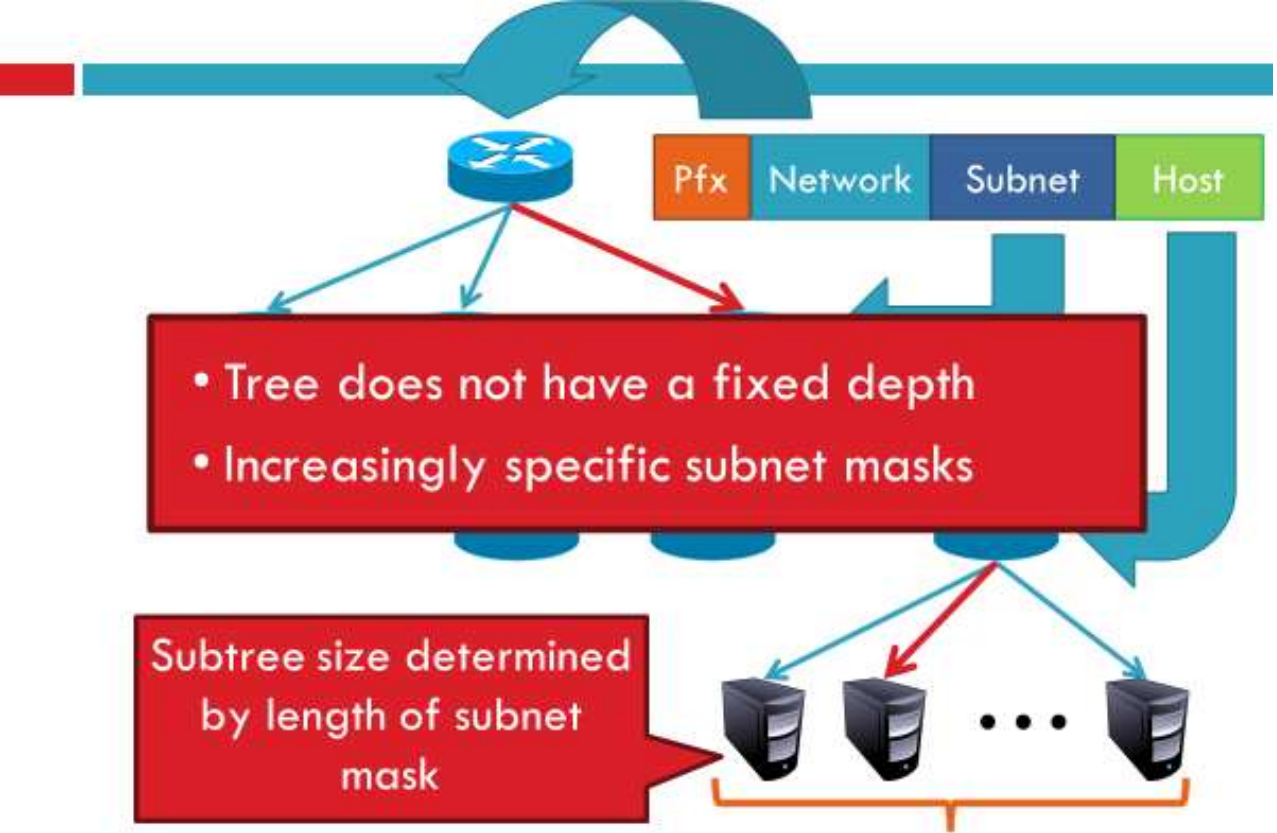
Subnet Mask: 11111111 11111111 11000000 00000000



Classless Addressing: Subnets

- ❖ Problem: Need to break up large A and B classes
- ❖ Solution: add another layer to the hierarchy
- From the outside, appears to be a single network
 - Only 1 entry in routing tables
- Internally, manage multiple subnetworks
 - Split the address range using a subnet mask

Classless Addressing: N-Level Subnet Hierarchy



Classless Addressing: Subnetting Examples



Thank You

